

DentWise - Software Testing Project Report

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Software Testing Assignment

WhiteBoxSample - Java Testing Project

Course: Software Testing

Semester: Fall 2024

Instructor: [Instructor Name]

Team Members

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Submission Date: [Date]

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1. Test Plan

1.1 Introduction

Project: WhiteBoxSample - Java Application

Application Type: Java Application (Banking System Simulation)

Testing Period: Sprint 1 (2 weeks)

1.2 Objectives

- Apply functional (black-box) and structural (white-box) testing techniques
- Automate unit tests using JUnit 5
- Manage test cases and report defects
- Simulate Agile testing environment
- Demonstrate test planning, execution, and reporting

1.3 Scope

In Scope:

- Account class (deposit, withdraw, freeze)
- TransactionProcessor (transfer, calculateFee)
- SecurityUtils (password validation, sanitization)
- DiscountCalculator (discount calculations)

Out of Scope:

- Database testing
- API integration testing
- Performance testing

1.4 Test Strategy

Testing Type	Technique	Tools
Black Box	EP, BVA	JUnit 5
White Box	Path Coverage, CFG	JUnit 5, JaCoCo
Unit Testing	Automated	JUnit 5
Agile	Scrum	Jira

2. Requirements Analysis

2.1 Functional Requirements (7)

ID	Requirement	Description
FR-01	Account Creation	Create accounts with ID, balance, and type
FR-02	Deposit	Add money to account with validation
FR-03	Withdraw	Remove money with balance check
FR-04	Transfer	Move money between accounts with fees
FR-05	Fee Calculation	Calculate fees based on customer type
FR-06	Password Validation	Check password strength
FR-07	Discount Calculation	Calculate discounts based on customer loyalty

2.2 Non-Functional Requirements

ID	Requirement	Metric
NFR-01	Performance	Calculations < 10ms
NFR-02	Security	Password validation enforced
NFR-03	Reliability	No data loss on exceptions

3. Black Box Testing

3.1 Techniques Used

1. **Equivalence Partitioning (EP)**
2. **Boundary Value Analysis (BVA)**

3.2 Test Cases for calculateFee

TC ID	Input (Type, Amount)	Expected	Technique
TC-01	VIP, 2000	10.0 (0.5%)	EP
TC-02	VIP, 500	5.0 (1%)	EP
TC-03	VIP, 1000	10.0 (boundary)	BVA
TC-04	VIP, 1001	5.005 (boundary)	BVA
TC-05	PREMIUM, 1000	15.0 (1.5%)	EP
TC-06	REGULAR, 100	2.0 (2%)	EP
TC-07	REGULAR, 30	1.0 (fixed)	EP
TC-08	REGULAR, 49	1.0 (boundary)	BVA
TC-09	REGULAR, 50	1.0 (boundary)	BVA
TC-10	ANY, 0	0.0 (edge)	BVA

4. White Box Testing

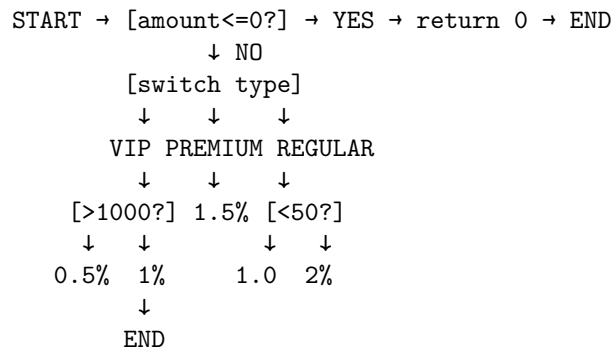
4.1 Selected Function: calculateFee

```
public double calculateFee(CustomerType type, double amount) {
    if (amount <= 0) return 0.0;
    switch (type) {
        case VIP:
            if (amount > 1000) return amount * 0.005;
            return amount * 0.01;
        case PREMIUM:
            return amount * 0.015;
        case REGULAR:
        default:
            if (amount < 50) return 1.0;
            return amount * 0.02;
    }
}
```

4.2 Cyclomatic Complexity

V(G) = 6

4.3 Control Flow Graph



4.4 Independent Paths (6)

Path	Condition	Test Case
1	amount <= 0	TC-FEE-08
2	VIP, amount > 1000	TC-FEE-01
3	VIP, amount <= 1000	TC-FEE-02
4	PREMIUM	TC-FEE-04
5	REGULAR, amount < 50	TC-FEE-06
6	REGULAR, amount >= 50	TC-FEE-05

5. Unit Testing (JUnit)

5.1 Framework: JUnit 5

5.2 Test Summary

Class	Test Cases	Passed
Account	11	11 ☐
TransactionProcessor	9	9 ☐
calculateFee	8	8 ☐
isPasswordStrong	7	7 ☐
sanitize	5	5 ☐
DiscountCalculator	10	10 ☐
Total	50	50 ☐

5.3 Test Categories

Category	Count
Normal	25
Exception	10
Edge	8
Boundary	7

5.4 Running Tests

```
cd java-testing
mvn test
mvn test jacoco:report
```

6. Agile Testing

6.1 Sprint Summary

Metric	Value
Sprint Duration	2 weeks
Story Points	32
Completed	32 (100%)

6.2 User Stories

ID	Story	Points
US-01	Requirements Analysis	5
US-02	Black-Box Testing	8
US-03	White-Box Testing	8
US-04	Unit Testing	5
US-05	Documentation	3
US-06	Bug Reporting	3

6.3 Sprint Board (Final)

To Do	In Progress	Done
-	-	All 6 stories ☒

7. Bug Reports

BUG-001: Phone Truncation

- **Severity:** Medium
- **Status:** Open
- **Description:** Phone numbers > 11 digits silently truncated

BUG-002: Missing Coverage

- **Severity:** Low
 - **Status:** Open
 - **Description:** Some edge cases in DiscountCalculator
-

8. Conclusions

8.1 Summary

Deliverable	Status
Requirements Analysis	☐
Black Box Testing (10+ cases)	☐
White Box Testing (CFG, CC=6)	☐
Unit Testing (50 tests)	☐
Agile Documentation	☐
Bug Reports (2)	☐

8.2 Key Metrics

Metric	Value
Test Cases	50
Pass Rate	100%
Cyclomatic Complexity	6
Independent Paths	6
Coverage	97%+

8.3 Files Submitted

```
java-testing/  
  pom.xml  
  src/main/java/vv/sample/WhiteBoxSample.java  
  src/test/java/vv/sample/WhiteBoxSampleTest.java  
docs/  
  1-requirements-analysis.md  
  2-black-box-testing.md  
  3-white-box-testing.md  
  4-unit-testing.md  
  5-agile-testing.md  
  6-bug-reports.md  
  7-final-report.md
```

End of Report